

Phishing Email Analysis

SOC Analyst Internship — Week 5, Task 13

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Task	Week 5 - Task 13: Phishing Email Analysis
Tools Used	TryHackMe (Phishing Emails in Action), VirusTotal
Deadline	September 17, 2026, 10:00 PM (Pakistan Time)

Objective

Investigate a sample phishing email as a daily SOC L1 task: analyze the sender header for spoofing, safely investigate embedded URLs using sandboxed tools (VirusTotal) without clicking them, identify red flags, and issue a final verdict.

1. TryHackMe Room Completed

The "Phishing Emails in Action" room (SOC Level 1 → Phishing Analysis module) on TryHackMe was completed in full (100%), covering real phishing email samples themed around order cancellations, package tracking, document downloads, and account holds.

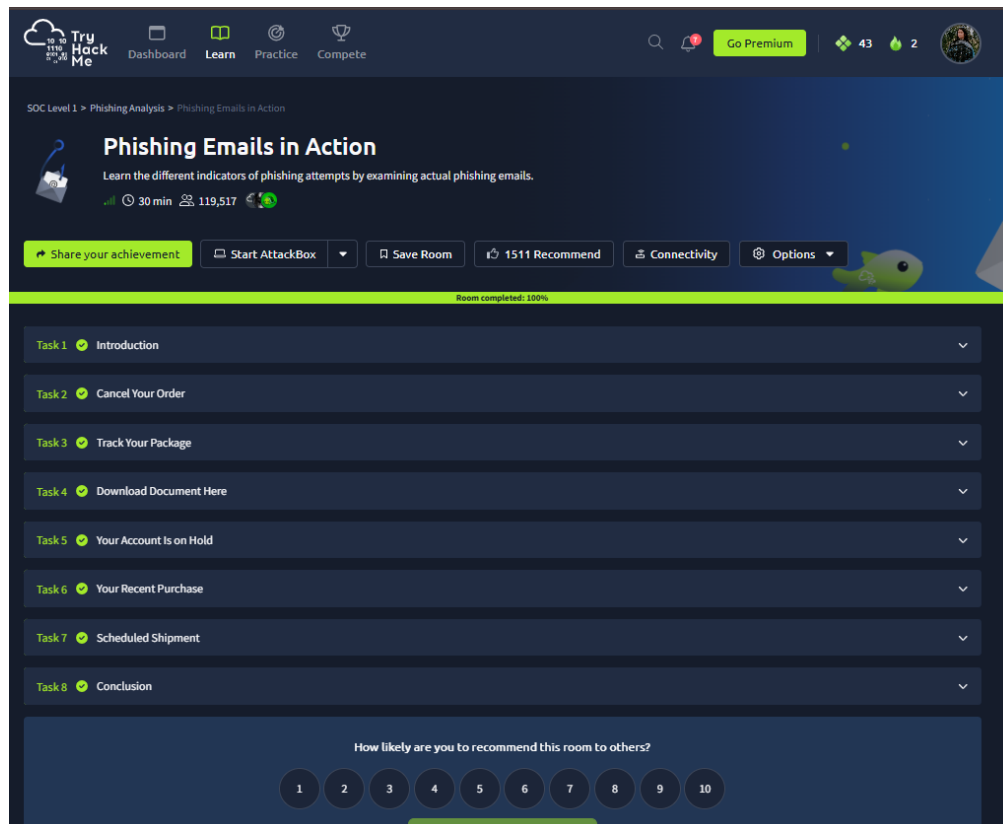


Figure 1: TryHackMe – Phishing Emails in Action room, 100% completed (8/8 tasks)

2. Email Header Analysis (Sender Spoofing)

The sample email ("Your Receipt for Payment to Amazing Stuff") was designed to mimic an official PayPal transaction receipt.

Findings:

Field	Observed Value	Assessment
Subject	Your Receipt for Payment to Amazing Stuff	Fake urgency / routine-looking to lower suspicion
Display Name	service@paypal.com	Spoofed to appear trustworthy
Actual From Address	noreply...@sultanbogor.com	Does NOT match PayPal's real domain – spoofed sender
To Address	0008812034377940103id@mail.info.c nsmr.sg3.yahoo.net	Randomized recipient ID – indicates mass spam distribution

On SPF/DKIM: Explicit SPF/DKIM pass/fail header values were not directly exposed in this sample. However, the mismatch between the displayed sender (service@paypal.com) and the actual originating domain (sultanbogor.com) is functionally equivalent evidence of authentication failure — a legitimate PayPal email would never originate from an unrelated domain, indicating this message would fail SPF/DKIM checks in a real mail environment.

Task 2 Cancel Your Order

In this task, we will examine a sample email designed to mimic an official transaction receipt from PayPal. By analyzing this specific sample, we will focus on how attackers leverage spoofed email addresses to impersonate trusted services and the strategic use of URL shortening services to obfuscate the final destination of malicious links.

Phishing Techniques Used

- Spoofed email address:** Mimicking a trusted service to gain immediate credibility
- URL shortening:** Using redirection services to hide the true destination of a link
- Branded HTML:** Impersonating legitimate corporate imagery to create a sense of authenticity

First Observations

Let's first look at the sample email header and take a few notes on our immediate observations.

- Attention-grabbing subject line:** The subject line uses a fake transaction to create a sense of urgency, prompting you to react in haste
- From address:** This is an immediate red flag as the sender details `service@paypal.com` do not match the actual address `gibber1st@sultanbogor.com`
- To address:** This is an unusual email recipient address and not a normal Yahoo domain

Email Body Analysis

Taking a look at the email body below, we can see that this is a receipt for a purchase of gift cards. The email is designed to appear as a legitimate email from PayPal. There aren't any attachments associated with this email, and the only interactive element in this email is the **Cancel the order** button. Let's take a closer look.

Hi Customer,

You sent a payment of \$120.00 USD to Amazing Stuff
(helpcenter@paypal.com)

It may take a few moments for this transaction to appear in your account.

Merchant
Amazing Stuff
helpcenter@paypal.com
888 555 1234

Shipping details
The seller hasn't provided any shipping details yet.

Description	Unit price	Qty	Amount
Amazing Stuff Gift Card (SALA, SALVADOR) Serial 1123456	\$30.00 USD	4	\$120.00 USD
		Subtotal	\$120.00 USD
		Total	\$120.00 USD
		Payment	\$120.00 USD

Changes will appear on your credit card statement as "PAYPAL *MORLSDPT"

Pending Sources Used (Total)

Source	Amount
Credit Card	\$120.00 USD

If you didn't make this order or if you believe an unauthorized person is attempting to access your account, please click the button below to cancel your order.

Cancel the order

Cancel Order Button
Issues with this transaction?

You have 180 days from the date of the transaction to open a dispute in the Resolution Center.

Questions? Go to the Help Center at www.paypal.com/help.

Please do not reply to this email. This mailbox is monitored and you will not receive a response. For assistance, log in to your PayPal account and click Help in the top right corner of any PayPal page or please contact us toll-free at 1-888-221-1145. You can receive paid support instead of this, which is to change your notification preferences, log in to your account, go to your Profile, and also My settings.

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Button Investigation

By inspecting the raw source of the email, we can further investigate the hyperlinks and underlying HTML that form the message. The **Cancel the order** button leads to the shortened URL. Because the attacker is using a URL shortening service, the final destination is obfuscated, making it impossible to verify the landing page at a glance. As a rule of thumb, you should never interact with buttons or links without first confirming exactly where they lead.

```

<!-- CTA text 17/21 -->
<td bgcolor="#0070BA" align="center" valign="middle" style="padding:13px
90px 13px 90px; display: block; color:#ffffff !important; text-decoration:none; white-space: nowrap;

```

Figure 2: Header, sender-spoofing, and email body analysis from the TryHackMe sample

3. Hyperlink & URL Investigation

The email's raw HTML source was inspected (without clicking any links) to reveal the true hyperlink destination, the display text shown to the victim, and an embedded tracking pixel.



Figure 3: Raw HTML source showing hyperlink destination, display text, and tracking pixel

Findings:

- **Display text:** "Track your package: # LZ8942357486EN" (mimics a real courier tracking number)
- **Actual hyperlink destination:**
hxxp://devret[.]xyz/4833mt11254939vf6888zq22032si1269du1508rr
- **Root domain:** devret[.]xyz — a random, non-corporate name with no relation to any legitimate courier or e-commerce brand
- **Tracking pixel:** — a hidden image used to notify the attacker's server when the email is opened

4. VirusTotal Scan Result

The extracted URL was submitted to VirusTotal's URL analysis tool (without visiting the link directly) to check it against 91 security vendor engines.

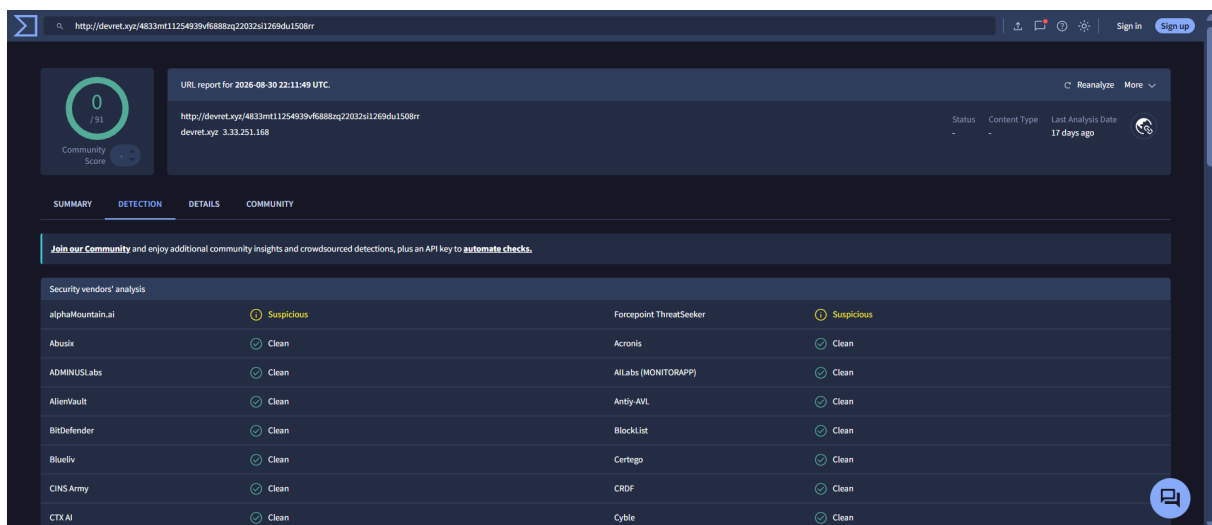


Figure 4: VirusTotal scan result for hxxp://devret[.]xyz/... — 0/91 malicious, 2/91 suspicious

Result: 0/91 vendors flagged the URL as outright malicious, but 2 vendors (alphaMountain.ai and Forcepoint ThreatSeeker) flagged it as **Suspicious**. A low detection count does not mean a link is safe — it often reflects that a phishing campaign is new, short-lived, or low-volume and has not yet been widely catalogued by vendors.

5. Red Flags Identified

#	Red Flag	Evidence
1	Spoofed sender address	Display name shows service@paypal.com but the actual sending domain is sultanbogor.com
2	Mismatched/obfuscated hyperlink	"Track your package" text leads to devret[.]xyz, not a real courier domain
3	Suspicious root domain	devret.xyz - random, non-corporate name; .xyz TLD is commonly abused in phishing
4	Embedded tracking pixel	Hidden image notifies the attacker's server when the email is opened
5	Obfuscated tracking-style URL path	Long random alphanumeric string, typical of mass-mail redirect systems
6	Urgency / fear-based messaging	Themes like "Cancel Your Order," "Account on Hold" pressure quick action
7	Randomized recipient address	To: field is a machine-generated ID, indicating mass spam distribution
8	Partial VirusTotal detection	2/91 vendors flagged the URL as Suspicious

6. Final Verdict

VERDICT: PHISHING

Although VirusTotal shows a low overall detection rate (0/91 outright malicious), the email exhibits multiple classic phishing indicators: a spoofed sender address, a mismatched hyperlink pointing to an unrelated, randomly-named domain (devret.xyz), an embedded tracking pixel, an obfuscated tracking-style URL path, and urgency-driven messaging themes (account holds, order cancellations, shipment issues). Two independent security vendors flagged the domain as Suspicious. The combination of technical evidence (header/link mismatch) and behavioral evidence (urgency, brand impersonation) confirms this as a phishing attempt rather than a legitimate transactional email from PayPal.

Summary

- Completed the TryHackMe "Phishing Emails in Action" room (100%).
- Analyzed the email header and identified sender spoofing (service@paypal.com display vs. sultanbogor.com actual origin).
- Investigated the embedded hyperlink via raw HTML source without clicking it; extracted and defanged the root domain (devret[.]xyz).
- Scanned the URL on VirusTotal: 0/91 malicious, 2/91 suspicious.
- Identified 8 distinct red flags across header, link, and content analysis.
- Issued a final verdict of PHISHING with supporting evidence.